Non-Solinas primes that give an SNFS-type route to `E(F_p)`: verified

pairing-friendly families Instrument: `examples/snfs_pairing_prime_families.rs` (run time under 2 s). Rows: `verified_families.tsv`. Figure: `figures/families.svg`. Conductor task T-5. Context: `research/elliptic_snfs_survey_20261016` §0 row A and §4A; `research/solinas_distribution_20261023`.

Answer

The only primes that open a special-number-field-sieve route to an elliptic curve are the primes of **pairing-friendly polynomial families**. They are not Solinas primes. The route is not on `E(F_p)` itself: it is the MOV / Frey–Rück map into `F_{p^k}*` with small embedding degree `k`, where the special polynomial `p = P(u)` with small coefficients makes `F_{p^k}` a target for SNFS-type algorithms (SexTNFS). A prime of any other shape, special or not, has `k` about as large as `r` for a curve of prime order, and the target field `F_{p^k}` is out of reach.

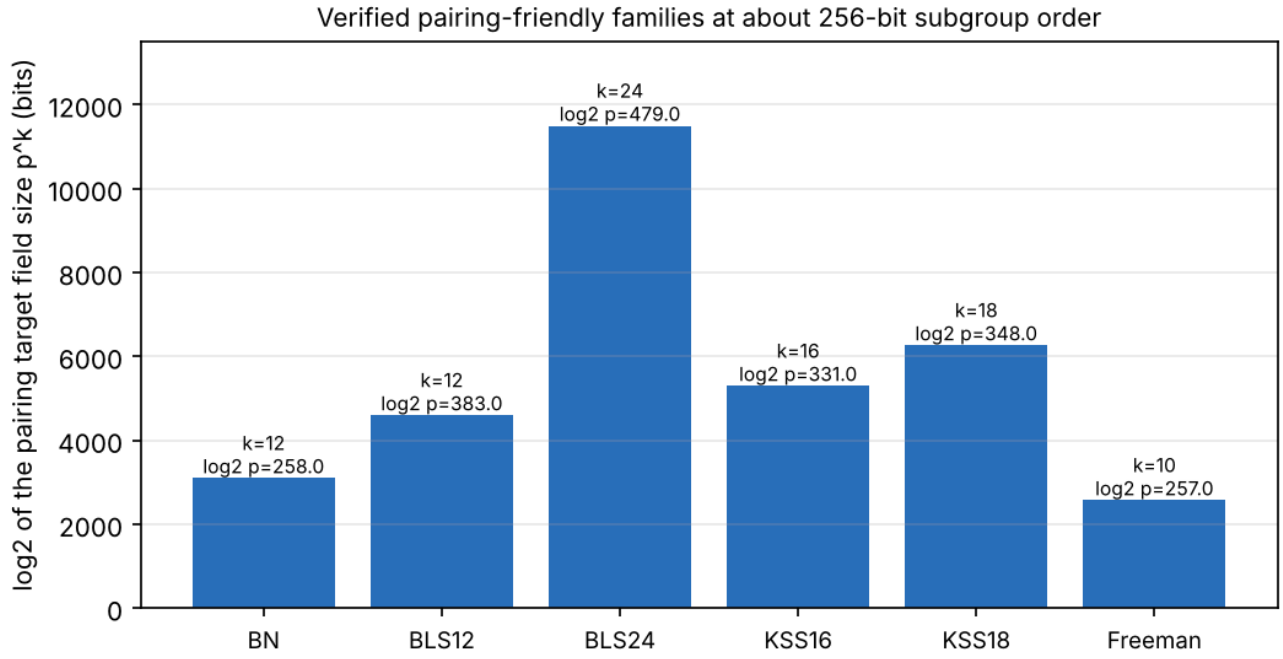
Claim classes: the rows below are **exact computations**. The statement that no other prime shape opens a route is the survey's literature-based argument (`[lit]`), not something this run establishes.

Verified parameters (about 256-bit subgroup order)

Each row was checked with exact big-integer arithmetic: `p` and `r` prime, `r | p + 1 − t` for the family's trace polynomial `t`, and the embedding degree `min{k : r | p^k − 1}` computed by the repository's `embedding_degree` equals the family's `k`. The program reports a failed check as a failure. All 12 rows passed. The first row of each family:

family	deg P	$\log_2$ p	$\log_2$ r	ρ = log p / log r	verified k	$\log_2$ p^k	u (hex)
Barreto–Naehrig	4	258	258	1.00	12	3096	0x8000000000000359
BLS12	6	383	257	1.49	12	4596	0x10000000000001cb6
BLS24	10	479	385	1.24	24	11496	0x100000000016e
KSS16	10	331	257	1.29	16	5296	0x40000ef75
KSS18	8	348	256	1.36	18	6264	0x1000000018fa
Freeman	4	257	257	1.00	10	2570	0x80000000000021fe

Polynomials used, with the family trace `t`: BN `p = 36u⁴+36u³+24u²+6u+1`, `r = 36u⁴+36u³+18u²+6u+1`, `t = 6u²+1`; BLS `r = Φ_k(u)`, `p = (u−1)²r/3 + u`, `t = u+1` (`k = 12, 24`); KSS16, KSS18 and Freeman as written in the example source. A second parameter set per family is in the table file.



Each row verified by exact arithmetic: p, r prime; r divides p+1-t; embedding degree equals k. Source: verified_families.tsv

What this does not show

- **No security level is claimed.** SexTNFS cost at these field sizes needs a verified cost formula. The BN-254 and BLS12-381 figures quoted in the survey come from secondary sources and stay marked unverified.
- A smaller ρ or a shorter `p` does not mean the prime is more exposed: the exposure is the size and special structure of `p^k`.
- These curves are pairing curves, not general ECDLP curves, and this is not an improvement over the index-calculus baseline or over matched rho.
- Primes of other special shapes (Proth/FFT primes `c·2^m + 1`, `Φ_d(m)` primes, smooth `p ± 1`) are special for `F_p*` or for the PKM factor base, which the repository measured as giving no end-to-end gain. They give no small `k`.
- Not covered: Cocks–Pinch and Brezing–Weng constructions, MNT curves (`k = 3, 4, 6`), and composite-degree towers.